ARTIFICIAL INTELLIGENCE - ASSIGNMENT 02

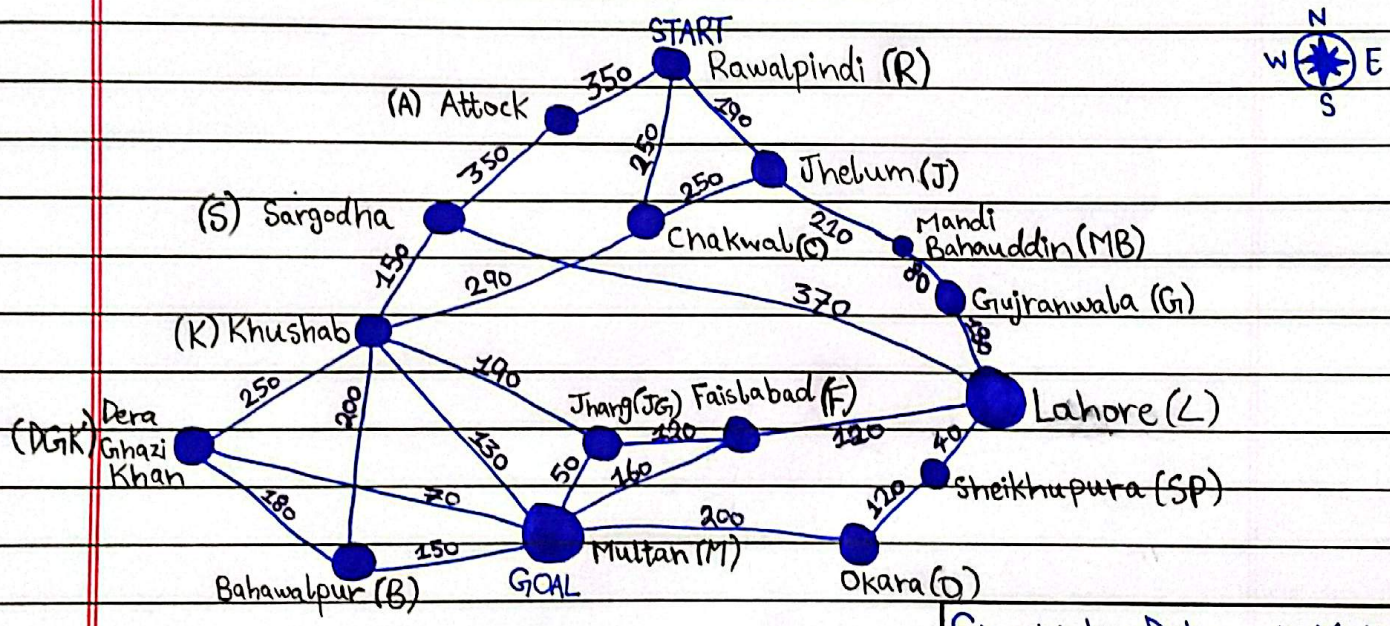
Date: 09/11/2026

NAME:-

Muhammad Hussain Durrani.

ROLL NO:-

L1F24BSCS0937



	Node	g(n)	h(n)	f(n)
1. (R)	Rawalpindi	0	350	0+350 = 350
2. (J)	Jhelum	190	200	190+200 = 390
3. (C)	Chakwal	250	150	250+150 = 400
4. (MB)	Mandi Bahauddin	400	60	400+60 = 460
5. (A)	Attock	350	120	350+120 = 470
6. (G)	Gujranwala	480	120	480+120 = 600
7. (K)	Khushab	540	120	540+120 = 660
8. (M)	Multan	670	0	670+0 = 670

Straight Line Distance to Multan (Hueristic)

Multan (M)	0	Sheikhupura (SP)	80
Bahawalpur (B)	70	Lahore (L)	180
Dera Ghazi Khan (DGK)	70	Gujranwala (G)	120
Khushab (K)	120	Mandi Bahauddin (MB)	60
Sargodha (S)	220	Chakwal (C)	150
Jhang (JG)	130	Jhelum (J)	200
Faisalabad (F)	85	Rawalpindi (R)	350
Okara (O)	180	Attock (A)	120

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PART A

(Q.NO.1)

-: A* Execution :-

Start = Rawalpindi
Goal = Multan
Evaluation Function:

$$f(n) = g(n) + h(n)$$

Where:

- g(n) = actual cost from start to current node
- h(n) = heuristic estimate to goal
- f(n) = total estimated cost

A* selects the node having the smallest f(n).

A* Search Trace

OPEN is ordered by increasing f(n). Each entry is written as City(f).

~~Step Expanded Node g(n) h(n) f(n)~~

	Frontier	Expanded	Explored
1.	J(390), C(400), A(470)	R(350)	R
2.	C(400), MB(460), A(470)	J(390)	R, J
3.	MB(460), A(470), K(660)	C(400)	R, J, C
4.	A(470), G(600), K(660)	MB(460)	R, J, C, MB
5.	G(600), K(660), S(920)	A(470)	R, J, C, MB, A
6.	K(660), L(840), S(920)	G(600)	R, J, C, MB, A, G
7.	M(670), B(810), L(840), JG(860), DG,K(860), S(920)	K(660)	R, J, C, MB, A, G, K
8.	//	M(670)	R, J, C, MB, A, G, K, M ; Goal Reached

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Final Path: Rawalpindi → Chakwal → Khushab → Multan
Total Path Cost = 250 + 290 + 130 = 670

Result: A* reaches Multan with a path cost of 670.

(Q.NO.2)

-: Heuristic Analysis :-

(a) Is the given heuristic admissible? Justify.

Ans:- No, the heuristic is not admissible.
A heuristic is admissible if:

$$h(n) \leq h^*(n)$$

where $h^*(n)$ is the actual minimum cost to reach the goal.
For Jhang:

$$h(JG) = 130$$
$$h^*(JG) = 50$$

Since $130 > 50$, the heuristic overestimates the actual cost.
Therefore, the given heuristic is 'not admissible.'

(b) Is the given heuristic 'consistent'? Justify

Ans:- No, the heuristic is not consistent.
A consistent heuristic most satisfy:

$$h(n) \leq c(n, n') + h(n')$$

For Jhang → Multan:

$$130 \leq 50 + 0$$
$$130 \leq 50 \quad \text{(False)}$$

Therefore, the heuristic is not consistent.

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(c) What happens if the heuristic overestimates?

Ans:- When a heuristic overestimates the remaining cost, A* may select a more expensive path and lose its guarantee of finding the optimal solution. It can still find an optimal path in some cases, but optimality is no longer guaranteed.

PART B

(Q.NO.3)

-: Water Jug Problem :-

Jug A capacity = 4 litres
Jug B capacity = 3 litres
Initial State = (0, 0)
Goal = exactly 2 litres in either jug

i) State Representation :-

A state is represented as:

$$(A, B)$$

where:

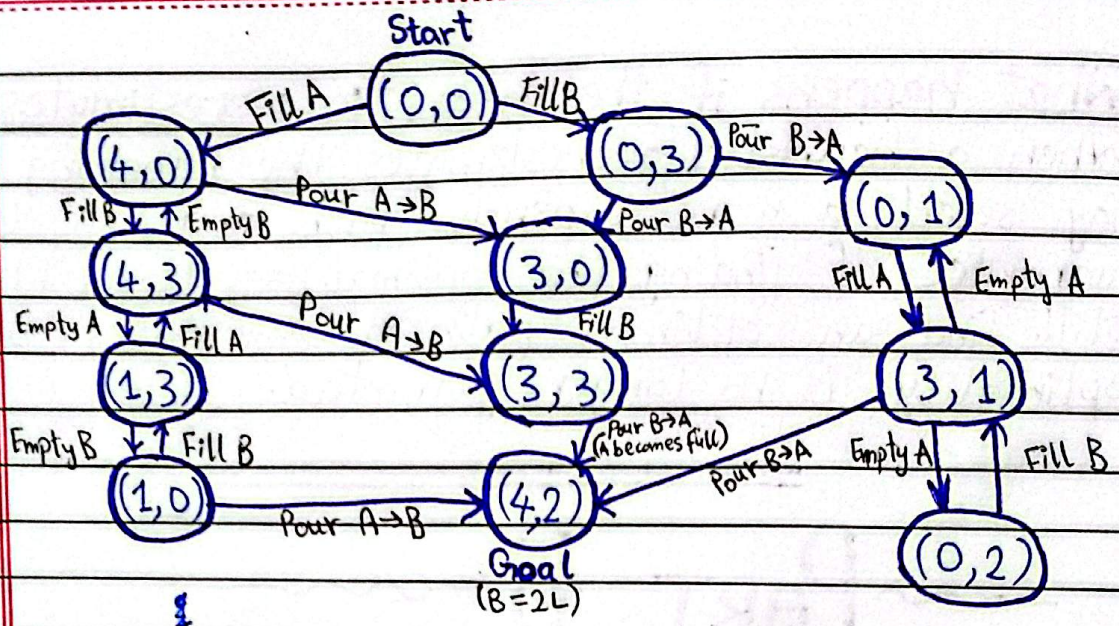
$$0 \leq A \leq 4 \quad , \quad 0 \leq B \leq 3$$

ii) Operators :-

a) Fill Jug A.
b) Fill Jug B.
c) Empty Jug A.

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d) Empty Jug B.
e) Pour water from A to B until A is empty or B is full.
f) Pour water from B to A until B is empty or A is full.

iii) BFS Search Trace:-

The frontier uses FIFO order, with the first state expanded first. Repeated states are not added again.

Step	Frontier (OPEN)	Expand	Explored (CLOSED)
0	(0,0)	—	∅
1	(0,3), (4,0)	(0,0)	(0,0)
2	(4,0), (3,0), (4,3)	(0,3)	(0,0), (0,3)
3	(3,0), (4,3), (1,3)	(4,0)	(0,0), (0,3), (4,0)
4	(4,3), (1,3), (3,3)	(3,0)	(0,0), (0,3), (4,0), (3,0)
5	(1,3), (3,3)	(4,3)	(0,0), (0,3), (4,0), (3,0), (4,3)
6	(3,3), (1,0)	(1,3)	(0,0), (0,3), (4,0), (3,0), (4,3), (1,3)
7	(1,0), (4,2)	(3,3)	(0,0), (0,3), (4,0), (3,0), (4,3), (1,3), (3,3)
8	(4,2), (0,1)	(1,0)	(0,0), (0,3), (4,0), (3,0), (4,3), (1,3), (3,3), (1,0)
9	(0,1)	(4,2)	..., ..., (4,2), Goal

iv) Final Solution Path:-

Step	Operation	State (A,B)
0	Both jugs empty	(0,0)
1	Fill jug B	(0,3)
2	Pour B into A	(3,0)
3	Fill Jug B again	(3,3)
4	Pour B into A until A is full	(4,2)

State-space Solution Path: (0,0) → (0,3) → (3,0) → (3,3) → (4,2)

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Result: Jug B contains exactly 2 litres.
Minimum number of operations: 4.

PART C

:- Critical Thinking :-

(Q.NO.4.)

:- Algorithm Behaviour :-

Path 1: Very deep but cheap.
Path 2: Shallow but expensive.

Algorithm	Preferred Path	Reason
• DFS	Depends on traversal order	DFS explores one branch deeply before backtracking
• UCS	Deep but cheap	UCS selects the path with the lowest accumulated cost $g(n)$
• Greedy Best-First	Depends on heuristic	Greedy selects the node with the smallest estimated remaining cost $h(n)$.
• A*	Deep but cheap, if its total cost is lower	A* uses $f(n) = g(n) + h(n)$. With an admissible heuristic and appropriate search conditions, it finds a minimum-cost solution.

Conclusion: UCS and A* prioritize total path cost rather than path depth: DFS is influenced by branch order, whereas Greedy depends on heuristic values.

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-: PART D :-

(Q.NO.5)

-: Debugging A* :-

Given incorrect function: $f(n) = h(n)$ only
Correct A function:* $f(n) = g(n) + h(n)$

- Explanation of the mistake: The student ignored $g(n)$, the actual cost from the start node to the current node. A* must consider both the accumulated path cost and the estimated remaining cost. Ignoring $g(n)$ can cause the algorithm to select a path that looks close to the goal but is expensive.

- What algorithm does it become?

It becomes Greedy Best-First Search, because it selects nodes using only the heuristic value $h(n)$.

-: PART E :-

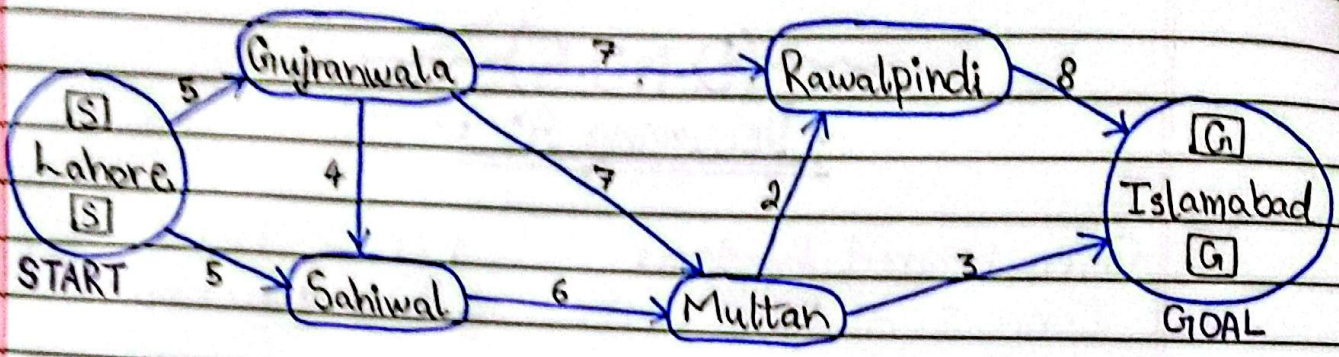
-: A* Search :-

Start: " Lahore
Goal: Islamabad
Evaluation functions: $f(n) = g(n) + h(n)$

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CITY CONNECTIVITY GRAPH (Punjab Region)



Heuristic Function [h(n)]

CITY (n)	h(n) [Straight-Line Distance To Islamabad]
Lahore	12
Gujranwala	8
Sahiwal	5
Rawalpindi	4
Multan	3
Islamabad	0

(a) A* Search Trace :-

Step	Frontier (OPEN),	Expand	Explored (CLOSED)
0	Lahore (12)	–	∅
1	Sahiwal(10), Gujranwala(13)	Lahore	L.
2	Gujranwala(13), Multan(14)	Sahiwal	L, S
3	Multan(14), Rawalpindi(16)	Gujranwala	L, S, G
4	Islamabad(14), Rawalpindi(16)	Multan	L, S, G, M
5	Rawalpindi(16)	Islamabad	L, S, G, M, I , Goal

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Evaluation Values:

Node	g(n)	h(n)	f(n)
Lahore	0	12	12
Sahiwal	5	5	10
Gujranwala	5	8	13
Multan	11	3	14
Islamabad	14	0	14

Final A* Path: Lahore → Sahiwal → Multan → Islamabad
Total Path Cost = 5+6+3 = 14

(b) BFS Search Trace :-

BFS expands nodes in FIFO order. The outgoing neighbors are considered in their visual top-to-bottom order in the provided graph.

Step	Frontier (OPEN)	Expand	Explored (CLOSED)
0	Lahore	—	∅
1	Gujranwala, Sahiwal	Lahore	L
2	Sahiwal, Rawalpindi, Multan	Gujranwala	L, G
3	Rawalpindi, Multan	Sahiwal	L, G, S
4	Multan, Islamabad	Rawalpindi	L, G, S, R
5	Islamabad	Multan	L, G, S, R, M
6	∅	Islamabad	L, G, S, R, M, I, Goal

Final BFS Path: Lahore → Gujranwala → Rawalpindi → Islamabad
Total Path Cost:

5 + 7 + 8 = **20**

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(c) Comparison of BFS and A* :-

Property	BFS	A*
Path	Lahore → Gujranwala → Rawalpindi → Islamabad	Lahore → Sahiwal → Multan → Islamabad
Total Cost	20	14
No. of Edges	3	3
Expansion Criterion	FIFO	Lowest g(n)+h(n)
Result	Minimum depth	Minimum cost for this graph

Conclusion: A* finds a lower-cost path (14) than BFS(20). BFS minimizes the number of edges, while A* considers both the accumulated cost and heuristic estimate.

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